

Hersita kumari

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EDUCATION

Trident Academy of Technology College of Engineering
Bachelor of Technology in Electronics and Telecommunications CGPA-7.94

Bhubaneswar, Odisha
2023 – 2027

TECHNICAL SKILLS

Languages: JavaScript, SQL, JAVA, Python
Frameworks: ReactJs, NodeJs, Tailwind, RestAPI
Developer Tools: Power BI, Google Cloud Platform, Postman
Database: MySQL, MongoDB
Version Control System: Git , GitHub
Relevant Courses: DBMS, Operating System, Data Structures, ETL, Data Analysis

EXPERIENCE

Software Development Intern
Tata Motors

May 2026 – Jul 2026
Jamshedpur, Jharkhand

- Developed and deployed an AI-powered Smart Kitting Verification System to automate the inspection of automotive component kits, reducing manual verification effort and improving inspection accuracy.
- Built a responsive React.js dashboard featuring live camera feed, reference kit upload, real-time component verification, inspection history, and progress reporting for seamless shop-floor monitoring.
- Integrated a Flask backend with YOLO-based object detection and OpenCV image processing to detect, classify, and validate kit components against predefined reference configurations in real time.
- Implemented automated kit comparison logic to identify missing, misplaced, and incorrect components, enabling faster quality assurance and minimizing assembly errors.
- Collaborated with industry mentors to optimize model performance through dataset annotation, confidence threshold tuning, image preprocessing, and UI enhancements for improved detection reliability.

PROJECTS

ML-Based ALU Synthesis (Cadence)

- Designed and synthesized an ALU using 180nm technology for high-performance computing applications.
- Utilized Cadence tools to evaluate configurations based on performance, power, and area.
- Applied machine learning concepts to automate design selection and optimize synthesis efficiency.

Zepyra: Real-Time Chat Application

- Built a real-time chat application using MERN stack and Socket.IO for instant messaging.
- Designed a responsive UI using TailwindCSS and improved performance with efficient state management.

[Link](#)

EXTRACURRICULAR

Hackathon — Smart Irrigation System

- Built an IoT-based system using ESP32-CAM, moisture and humidity sensors to automate irrigation based on plant health.
- Developed app/web interface for monitoring and control of water supply.

IoT Club — Mobibin Project

- Developed a smart dustbin with autonomous movement and obstacle detection using sensors.
- Integrated camera module and documented system design and implementation.

Internal Complaints Committee

- Handled student safety and misconduct cases with confidentiality and professionalism.